Reproducing A-Mem

Agentic memory for LLM agents, on a student budget

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Code: github.com/YUTING-FU-123/FTEC5660-IP2-amem-repro

What the system does

A-Mem (NeurIPS 2025) replaces flat memory with Zettelkasten notes.

- **Note construction** — keywords, tags, context for every utterance
- **Link generation** — connect to nearest notes
- **Memory evolution** — rewrite neighbours' context and tags
- **Retrieval** — top-k notes → answer

Multi-step, tool-calling, self-modifying → an agentic system, not a single prompt.

What we set out to reproduce

- Per-category **token-level F1 / BLEU-1** on LoCoMo
- Reported context cost: **2,520 tokens** per question (vs 16,910 full-context)
- Subset: **3 of 10 conversations** — 1,297 utterances, 500 questions

Category	Paper F1
Multi-hop	27.02
Temporal	45.85
Open-domain	12.14
Single-hop	44.65

Setup: everything is CPU + a cheap API

- MacBook Air, 8 GB, no GPU
- Py 3.12, MiniLM encoder, $k = 10$
- **Model swapped**: GPT-4o-mini → **DeepSeek V4.1 Flash**

4096

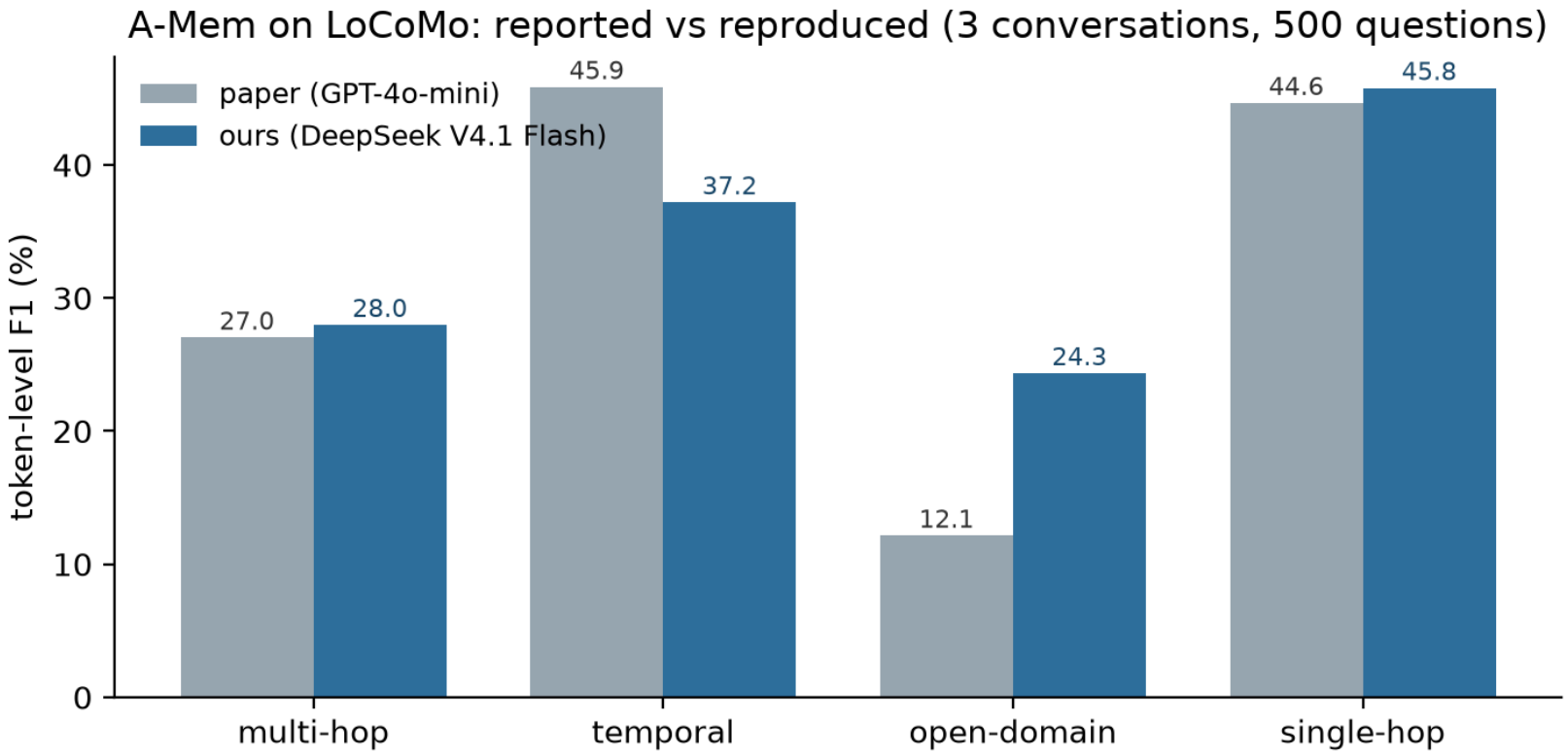
max_tokens — at 1000 the reasoning trace ate the budget and 23% of answers came back empty

0.6 s

per call with thinking disabled, vs 8.8 s with it on (1,213 → 219 completion tokens)

Recorded as controlled variables. Results are **not directly comparable** to the paper.

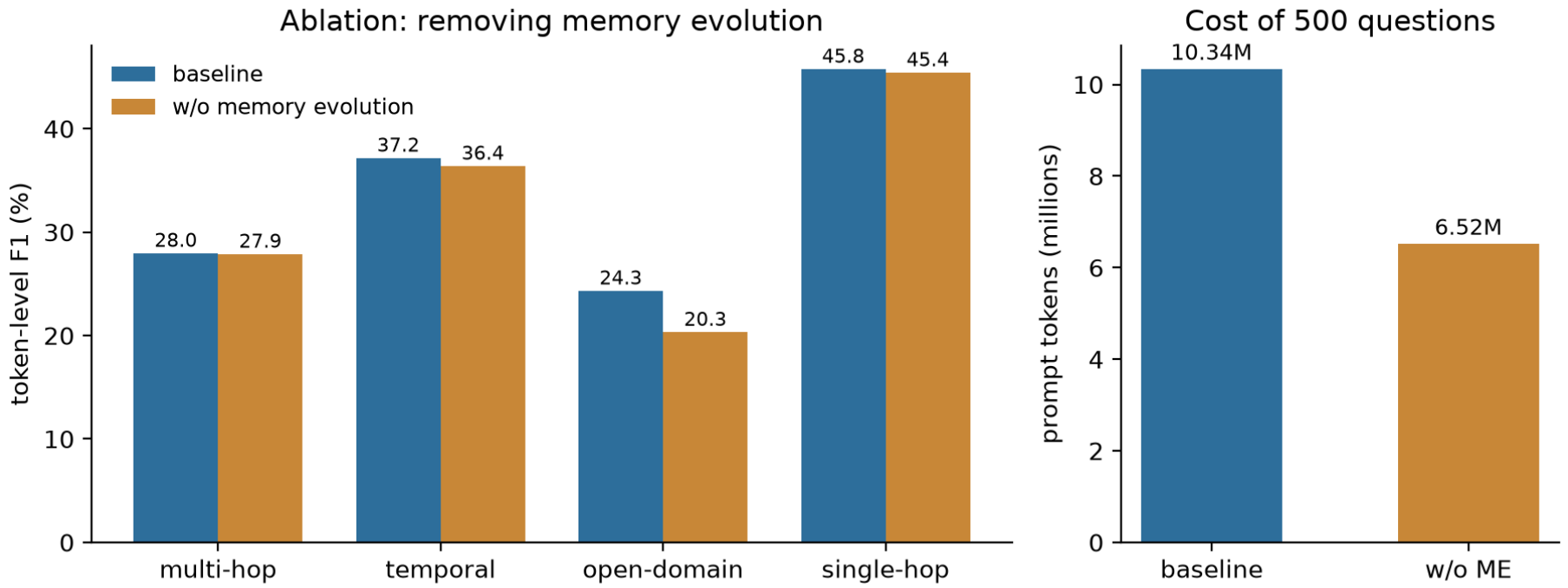
Results: two categories reproduce, two do not



Why the gaps are there

- **Temporal –8.7**: date normalisation, not retrieval — gold “19 January, 2023”, model “Yesterday, 20 January 2023”.
- **Open-domain +12**: we feed ~14.2k tokens of context per question, the paper reports 2,520.
- **Context cost does not reproduce**: 5.6× the reported number at the repo's default *k*.
- **Adversarial not comparable**: harness default scores against “Not mentioned”, the paper scored against the adversarial answer.

Our modification: remove memory evolution



Δ F1 = +0.02 · prompt tokens –37% · context per question –45%

Paper reports the opposite: multi-hop 27.02 → 21.35 without ME.

Lessons

- pitfall The harness caches memories **by sample index** — two subset files silently share sample 0. Tag the cache by dataset.
- cost One conversation is already ~1,700 LLM calls; measure tokens before choosing a scale.
- model swap Disable thinking and raise max_tokens, or 20% of outputs come back empty.
- finding Accuracy reproduces within ~1 point; **efficiency and ablation claims do not**.

Report · github.com/YUTING-FU-123/FTEC5660-IP2-amem-repro · raw JSON in `repro/out/`